

Vijay Kumar Pasala

Kadapa, Andhra Pradesh | pasalavijaykumar04@gmail.com | +91 79930 72927 | [LinkedIn](#) | [GitHub](#)

EXPERIENCE

Research Intern — IIT Tirupati Jan 2026 — Jul 2026
Robotics & Embedded Systems Lab, IISER-ISRO Collaboration

- Worked hands-on with an nRF-based embedded board, debugging hardware and firmware issues during bring-up and integration.
- Worked on real-time control and integration of a JAKA Zu5 6-DOF collaborative robotic arm for lab research experiments.
- Built a Random Position Machine (RPM) — a microgravity simulation system — integrating embedded hardware with control software over a TCP/IP-based communication layer.

Embedded Engineer — GlassData Present
Ambient Sensing / On-Device ML

- Developing embedded C firmware for on-device deployment of wake-word detection models on an ambient sensing board.
- Implementing signal preprocessing stages and iterating on model accuracy through data-driven testing across multiple hardware flashing cycles.
- Designed a master-slave UART communication protocol between the ambient sensing board and a companion controller for synchronized data exchange.
- Prototyping a vision-based detection pipeline as an adaptable secondary sensing modality alongside the audio path.

PROJECTS

Collaborative Robotic Arm Control System 2026
C/C++, Python, ROS2, OpenCV, IP Networking

- Built a client application enabling a single-click switch between a Gazebo-simulated environment and a real 6-DOF robotic arm connected over TCP/IP.
- Integrated the JAKA SDK and the underlying network communication layer between control software and hardware.
- Added an OpenCV-based computer-vision safety layer that monitors the workspace in real time and interrupts execution when a person enters it, achieving sub-second response latency validated through repeated trial testing.

Object Localization using ZED 2i Depth Camera 2026
Python, ZED 2i, DINO, Robot Kinematics

- Found the X, Y, Z positions of objects in the scene using depth data from a ZED 2i stereo camera.
- Organized the coordinate system with the robot base and computed the exact distance from the robot base to each object by matching depth values with pixel coordinates.
- Used DINO models for object detection within the pipeline.

LifeBand MAA — Health Monitoring 2025 — Present
Embedded C, Signal Processing

- Built a wearable ECG and heart-rate monitoring system performing real-time signal acquisition, filtering, and anomaly detection on streaming physiological data.
- Served results through a lightweight on-device inference pipeline.

EyeNtra 2025
AI-Powered Strabismus Diagnostic System — OpenCV, Flutter, Real-Time Image Processing

- Built an OpenCV-based computer-vision pipeline to detect strabismus (eye misalignment) by computing corneal reflex and pupil-tracking metrics with sub-millimeter precision from live medical image streams, exposed via REST API endpoints.
- Built a customized Flutter app as the diagnostic front-end for clinicians to capture eye images and view real-time screening results.
- Conducted practical testing at Arvind Eye Hospital, collecting patient data; testing and refinement of EyeNtra are ongoing.

Wireless Vibration Monitoring System — nRF5340 2026
BLE, Embedded Systems

- Implemented a 1 kHz real-time data acquisition system on the nRF5340.
- Designed a custom BLE GATT service for efficient high-frequency wireless data streaming.
- Optimized firmware for low-latency, reliable wireless communication under continuous sensor load.

Electronics and Communication Engineering undergraduate with hands-on experience in real-time embedded systems, closed-loop control, and applied signal processing (encoder/IMU/ECG data pipelines, PID tracking loops, sensor-fusion filtering). Strong C/C++ and Python fundamentals with working knowledge of computer networks, IP-based system integration, and Linux platforms.

SKILLS

Programming:
C, C++, Python, JavaScript

Signal & Systems:
Real-Time Control Loops, PID / Tracking Loops, Signal Filtering & Anomaly Detection, Sensor Fusion, Encoder/IMU Data Processing

Networking & Protocols:
TCP/IP, HTTP, REST APIs, BLE/GATT, UART (Master-Slave Communication), SPI, I2C

Embedded Platforms & Boards:
Arduino, ESP32, Raspberry Pi, nRF5340, Teensy, Ambient-Sensing Boards

Computer Vision & Simulation:
OpenCV, Object Detection, Contour Extraction, Gazebo, NVIDIA Isaac Sim

Embedded & Real-Time:
Embedded C, ROS2, Multithreading, Firmware Deployment, GPIO, Interrupts, PWM, Timers

Core Electronics:
Digital Electronics, Analog Circuits, Microcontrollers, ADC/DAC Concepts

Tools & Platforms:
Linux, Git, Docker, VS Code

EDUCATION

RGUKT RK Valley
B.Tech Electronics & Communication
2023 — 2027 | CGPA: 8.0

RGUKT RK Valley
Pre-University Course (PUC)
2021 — 2023 | CGPA: 9.63

WORKSHOPS & TRAINING

5G Hands-on Workshop — IIT Tirupati

- Deployed a complete 5G network from scratch using OpenAirInterface, containerized with Docker.
- Configured the 5G Core Network and integrated User Equipment (UE) with gNodeB for real-world simulation.
- Analyzed network performance, throughput, and dynamic bandwidth allocation across connected devices.

ACHIEVEMENTS

Hack the Flame — National Winner
LifeBand MAA wearable health system

Astronics 6.0 — College Winner
Obstacle avoidance robotics project

Peekuthon — Shortlisted
LifeBand MAA, top national entries